

I/O Buses & Magnetic Disk

1. What is the primary function of a bus in a computer system?
 - a) To store data permanently
 - b) To provide a communication path for data transfer.
 - c) To process instructions
 - d) To control memory allocation
2. Which of the following factors affects disc access time?
 - a) Seek time
 - b) Rotational latency
 - c) Data transfer rate
 - d) All of the above
3. A disk has each track with 1k sectors each with 4KB capacity and it takes 10msec for 1 rotation. The transfer rate of the disk is?
 - a) 300 MB/sec
 - b) 400 MB/sec
 - c) 450 MB/sec
 - d) 350 MB/sec
4. A magnetic disk has 5,000 tracks. The number of sectors per track is 100, and the size of each sector is 512 bytes. The disk has 10 platters, and each platter has two sides. What is the storage capacity of the disk in bytes? (GATE 2016 - CS)
 - a) 5,12,00,00,000 B
 - b) 25,60,00,00,00 B
 - c) 5,00,00,000 B
 - d) Not enough information
5. What synchronizes all devices in a synchronous bus?
 - a) A handshake signal
 - b) A common clock
 - c) A unique device address
 - d) A processor interrupt
6. In a synchronous bus, what is the role of the bus master?
 - a) It initiates data transfer by issuing read or write commands
 - b) It only receives data from the slave
 - c) It acts as a backup memory device
 - d) It prevents bus conflicts and allows a better communication
7. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete

rotation. The total time (in milliseconds) needed to read the entire file is _____
(GATE 2009-CS)

A) 14020

B) 14000

C) 25030

D) 15000

Ans. Total time needed to read entire file = $2000(4\text{ms} + 6\text{ms}/2 + 6\text{ms}/600)$
= 14200 milliseconds

8. Consider a hard disk with 36 recording surfaces (0-35) having 10000 cylinders (0-9999) and each track contains 64 sectors (0-63). Data in disk are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file in the disk is stored starting from address <1660, 28, 38>. What is the sector number of the first sector and last sector of file, if 55788 sectors stored in contiguous manner in the file in disk?

A) 382640, 438428

B) 382490, 453242

C) 382640, 453242

D) 382490, 438428

Ans. Sector number of the first sector of file = $1600 \times 2304 + 64 \times 28 + 38$
= 382640

Sector number of last sector of file = $382640 + 55788$
= 438428

9. Consider a disk with 16 platters, 2 surfaces per platter, 1K tracks per surface, 2K sectors per track and 2048 Bytes per sector. Disk rotates with 3000 rpm. Seek time is 10ms.

If the disk is used in burst mode of DMA, such that whenever (64-bits word is available, it will be transferred in 16ns. What is the % of time CPU is blocked ?

A) 28.0057%

B) 28.57%

C) 27.9%

D) 20%

Ans. For 8 bytes, time taken to transfer = 16ns

For 2048 bytes, time taken to prepare = 20×10^6 ns

For 8 bytes, time taken to prepare = 40 ns

% time CPU blocked in burst mode of DMA = $16 / (16 + 40) \times 100\%$
= 28.57%

10. For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to:

A) Non-uniform distribution of requests

B) Arm starting and stopping inertia

C) Higher capacity of tracks on the periphery of the platter

D) Use of unfair arm scheduling policies

11. Which of the following is a characteristic of magnetic disks?

- A) Sequential access
- B) Direct access
- C) volatile storage
- D) non-writable storage

12. A hard disk rotates at 7200 RPM. What is its average rotational latency? (GATE CSE 2012)

- A) 2 ms
- B) 4.17 ms
- C) 8.33 ms
- D) 12.5 ms

13. In an asynchronous bus, how does data transfer begin? (GATE CSE 2017)

- A) When the clock signal is received
- B) When the CPU polls the device
- C) When the sender sends a request and receives an acknowledgment
- D) When the device interrupts the CPU

14. The key difference between synchronous and asynchronous buses is: (GATE CSE 2017)

- A) Synchronous buses use handshaking, asynchronous buses use a clock
- B) Synchronous buses use a clock, asynchronous buses use Handshaking
- C) Synchronous buses are slower than asynchronous buses
- D) Asynchronous buses require a clock signal for operation

15. In an asynchronous bus, how is coordination between sender and receiver achieved? (GATE CSE 2014)

- A) Using a clock signal
- B) Using handshaking signals
- C) Using polling
- D) Using interrupts

16. Which of the following factors has the most impact on disk access time? (GATE CSE 2019)

- A) Disk cache size
- B) Seek time
- C) Data transfer rate
- D) Disk size